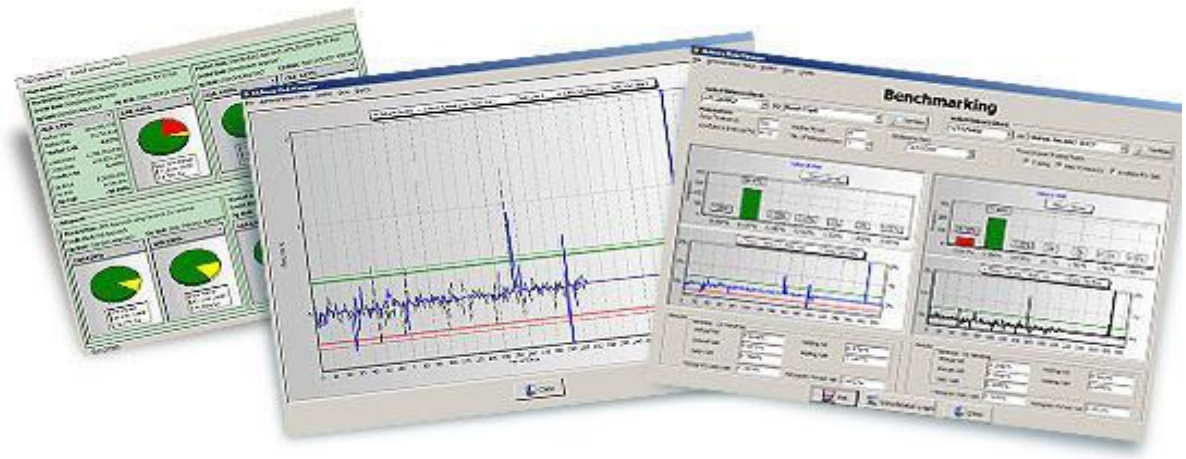
JS Bank

InfoBits STP – RTGS Gateway Application

Solution Deployment – Revised Proposal

March 10, 2016



204-205, Marine Faisal main Shahrah-e-Faisal, Block-6 PECHS Karachi Pakistan

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INFOBITS - INTRODUCTION

10 Years of Excellence in Software Development and Consulting Services.
Infobits was formed in 2002. In last decade of IT services, we have developed state of the art business application for diverse industries. Our products suite comprises;

- FRS - Financial Reporting System
- AMS - Assets Management System
- PPS - Production Planning System
- IMS - Inventory Management System
- HRPS- Human Resources & Payroll System
- HMS - Hospital Management System

After spending six years in business software application development, we decided to take things to the next level. In 2009 we started our Analytics and Business Intelligence division. Since then Infobits providing the right blend of industry knowledge and experience, helping organizations pave the way towards success.

Last couple of years, Infobits has been at the leading edge of developing custom business intelligence solutions leveraging our consultants' deep knowledge of business analytics, data warehousing, business intelligence and performance management using world class tools like **SAS, Business Object and Microsoft BI Suite**– to help a wide range of companies improve their core business processes and increase profitability. Our Analytics department providing following professional services in **Pakistan** and **Turkey**

- Project Management
- Business Analytics and Intelligence (BI)
- Customer Segmentation Modeling, objective prioritization, KPI generation, results measurement and strategic business planning.
- Credit risk scoring and rating
- Strategic Reporting and Analysis like Calculation of Customer LTV (Life Time Value), Monitoring Promotional Intensity and Expected Profit Calculations

Our Partners

ASRYM USA

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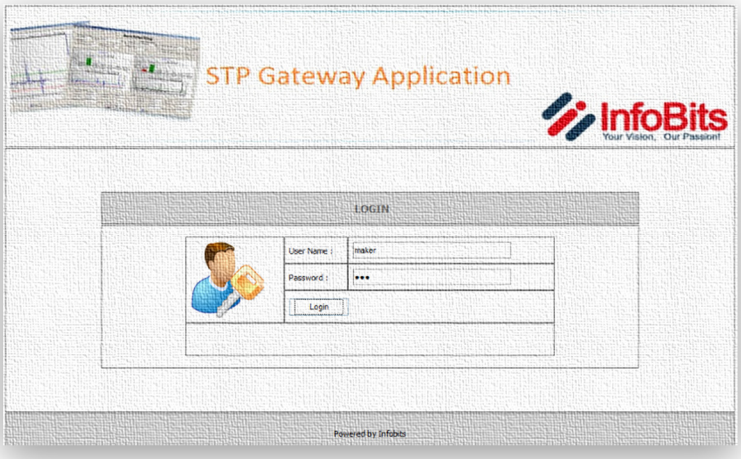
SOLUTION - INTRODUCTION

InfoBits STP - RTGS Gateway System provides the integration between Local banking software running at Bank's site and "Straight through Processing (STP)" running at Central Bank. It is the gateway between Core Banking System and Central Bank.

Since the inception of RTGS in Pakistan, enrollment of direct participants is growing and volume of transactions is also increasing. It has been foreseen that in future, volume of transaction will increase dramatically. This growing volume forcing banks to deploy a gateway application between Core Banking System and Central Bank to improving performance of financial messages communications channels.

The requirement is initiated by JS Bank that a gateway application system between Core banking and State Bank of Pakistan which will act as local STP RTGS solution for MT103/MT102/MT202 messages. It will also be a repository of messages exchanged between State Bank of Pakistan and JS Bank and provide reports for audit and reconciliation. There are three modules in InfoBits RTGS Gateway Application

1. Outward message routing
2. Inward message routing
3. Reporting module



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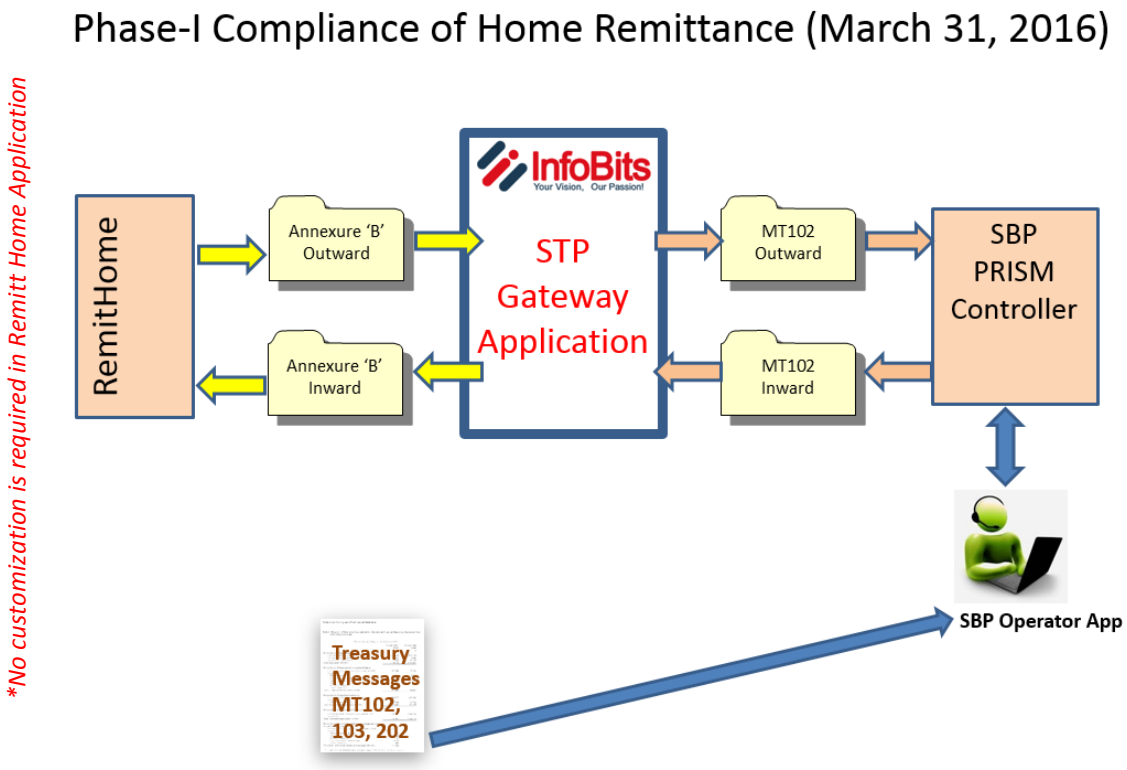
SCOPE OF DEVELOPMENT AGREEMENT

Software Technical development/deployment agreement is ("Agreement") made between **INFOBITS** and **JS BANK** ("Customer") for the RTGS Gateway application (Software application). Under this agreement INFOBITS hereby agrees to design and develop the Product in accordance with the following project outline as per the given scope:

Methodology

InfoBits STP-RTGS Gateway Application is centralize repository for a bank and State Bank of Pakistan for outward and inward messages. It is the gateway between Core Banking Application and State Bank of Pakistan's PRISM; the application will pull and push messages from Core Banking Application and State Bank of Pakistan's messages folders initially through "SBP Controller Application" and after deployment of InfoBits Operator Application, RCCC API communication channel will deployed and used in replacement of SBP Controller Application.

Overall Straight through Processing (STP) Project has been divided into three Phased



Phase-I Compliance of Home Remittance (March 31, 2016)

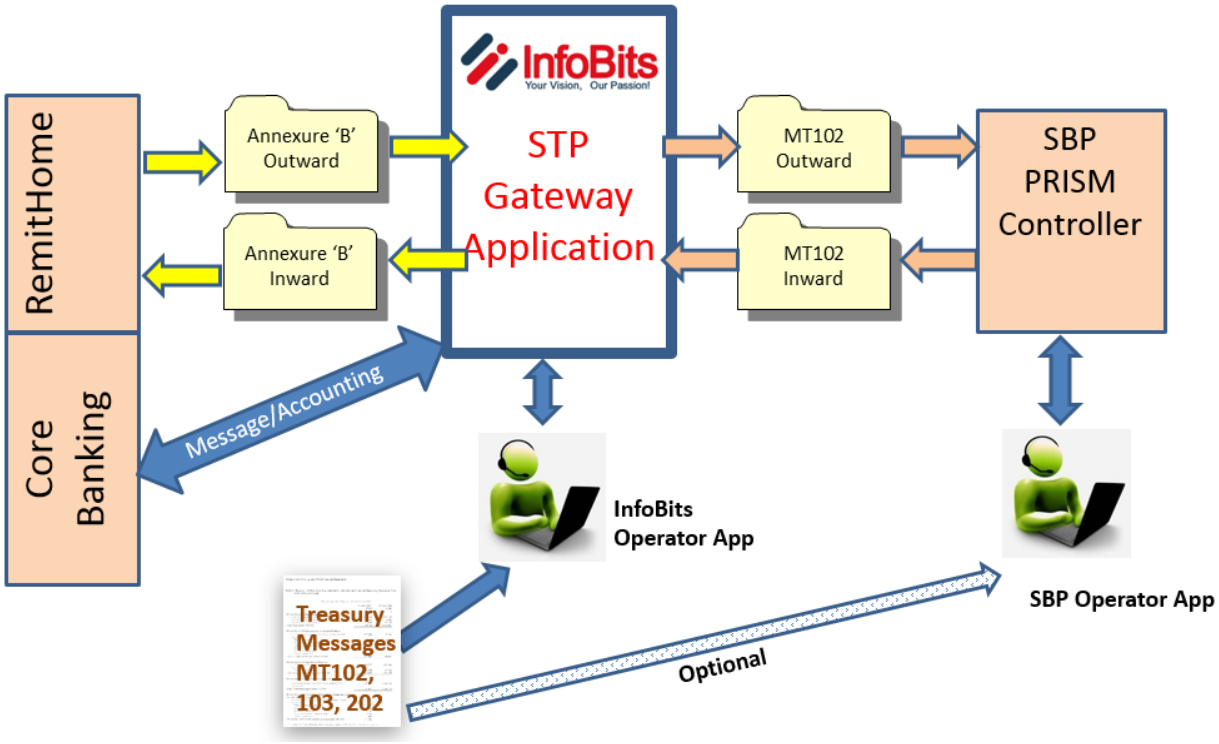
Outward

- Remit Home Application user will export all outward messages and will be placed in “Out RemitHome Folder” manually
- Message in annexure “B” format, will extracted by “InfoBits Gateway App” from “Out RemitHome Folder” and converted into MT102 format and will be place in “Outward Message Folder” accessible to “SBP Controller Application” automatically.
- “SBP Controller Application” user will import all MT102 messages from “Outward Message Folder” and transmit the message to SBP RTGS system using “SBP Controller Application”

Inward

- “SBP Controller Application” user will export all incoming MT102 messages from “SBP Controller Application” and will placed into “Inward Message Folder” using “SBP Controller Application – Export Option” manually.
- InfoBits Gateway App will extracted all inward MT102 messages from “Inward Message Folder” and will convert into annexure “B” format and will be place all those messages into “Inward RemitHome Folder” automatically.
- Through business as usual, Remit Home Application user will import all inward messages available in “In RemitHome Folder” manually.

Phase-II Deployment of Message Creation Application
(InfoBits Operator App)

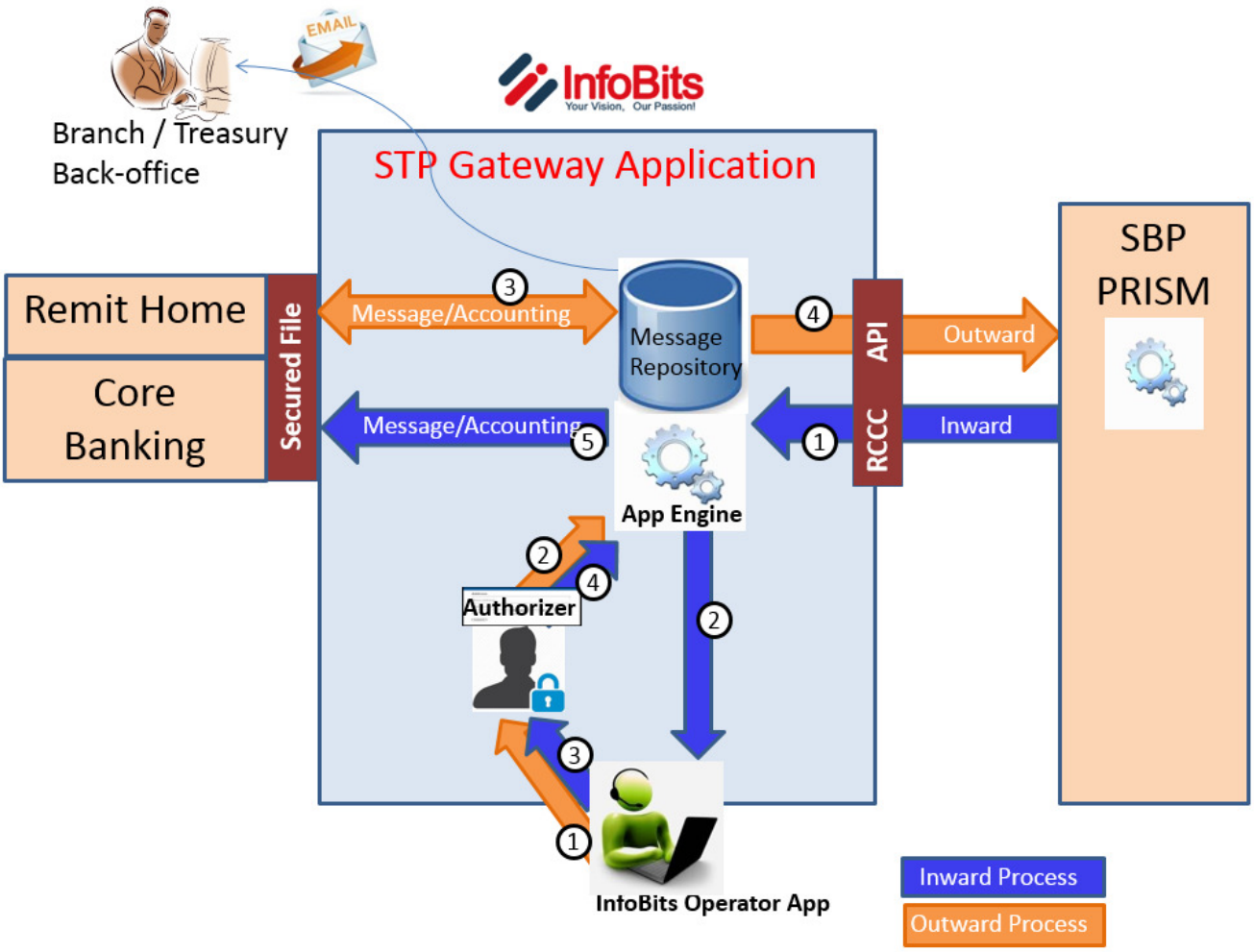


Phase – II Deployment of Message Creation Application

In this phase following application modules will be deployed and tested;

- InfoBits Integration Engine to integrate Core Banking Application with RTGS Gateway Application will be deployed and tested for Inward and Outward messages.
- Financial/Accounting entries posting mechanism will also be developed and deployed according to the requirement narrated by JS Bank.
- Deployment of InfoBits Controller Application for message creation will also be the part of this phase.

Phase – III Deployment of RCC Communication Channel in replacement of SBP Controller (Project Finalization)



Phase III will be targeting to make JS Bank fully compliant for Straight Through Processing (STP) with PRISM in accordance of given State Bank's guidelines.

Phase – III

Following End-To-End STP approach will be adopted for Inward message routing and outward message routing

Inward message routing

MT103, MT202 and MT202

(1) InfoBits RTGS Gateway application will collect financial messages from Central bank's PRISM System's message repository through RCCC API data stream communication channel.

(2) All financial messages will be placed in processing queue for Operator to validated beneficiary account number and account title from core banking application database (*through core banking API*).

(3) Once the account information provided in inward financial message is validated from core banking application database, operator will process the message for authorization.

(4) Authorizer will review the message and authorize the transaction for further processing.

(5) All authorized transactions will posted in the system and accounting will be performed in core banking application (*through core banking API*).

InfoBits STP-RTGS Gateway application will also send email intimation for all incoming messages to respective branch or group along with original message attached automatically.

Outward message

MT103, MT202 and MT 202

(1) All financial messages will be prepared on InfoBits RTGS Gateway application's operator interface. Once the message is prepared, it will be send to authorization queue.

(2) Authorizer will review the messages and authorize the transactions for further processing.

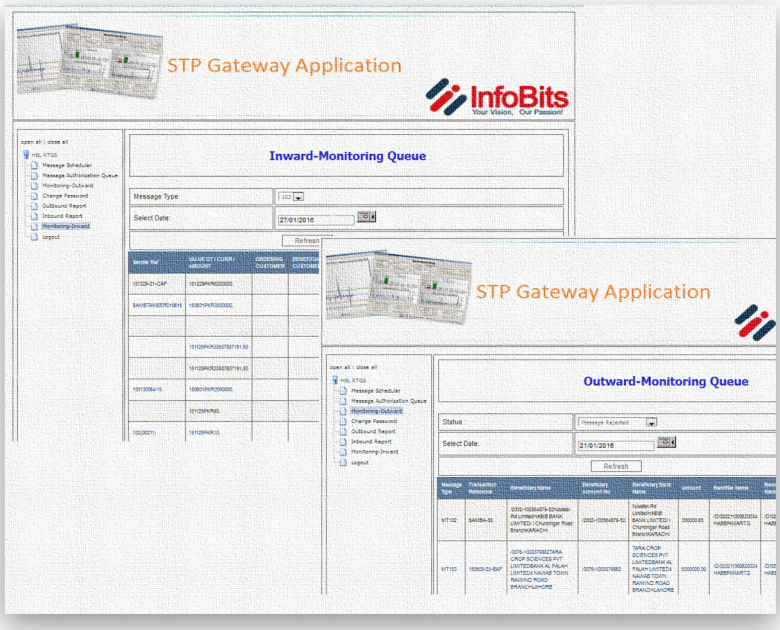
(3) All authorized transactions will posted in the system and accounting will be performed in core banking application database (*through core banking API*).

(4) Once accounting is performed successfully in core banking application database (*through core banking API*), the transactions will be transmitted to Sate Bank's RTGS System (PRISM) through RCCC API data stream communication channel.

InfoBits RTGS Gateway application will also send email intimation for all successful/Unsuccessful messages delivery report to respective branch or group along with original message attached automatically.

Online monitoring queue

Online monitoring queue for all incoming and outgoing messages, will be available with a status-wise drop down list



Message Queue

InfoBits RTGS Gateway Application removes the unprocessed messages from queue after cut-off time specified by JS Bank and generate a report. Intimation will also be generated for such removal via e mail. Records of all expired messages will be available in the system for future reference only.

Parameterized scheduler

User can set message cutoff time in application. Application will sent message to Central Bank's message repository as per define cutoff time and after that application will remove the message from the transfer queue.

Reporting module

Comprehensive reporting module is also available for printing message confirmations; acknowledgements to their respective department/Branches/user groups.

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1. *Outward Route message detail*

Report fields are mention below.

a. Transaction Reference

b. Beneficiary Name

c. Beneficiary Account No.

d. Beneficiary Bank Name

e. Amount

f. Remitter Name

g. Remitting Bank Name

h. Time of lodgment

i. Status

2. *Inward Route message detail*

Report fields are mention below.

a. Transaction Reference

b. Beneficiary Name

c. Beneficiary's Account No.

d. Amount

e. Remitter Name

f. Sender Bank Name

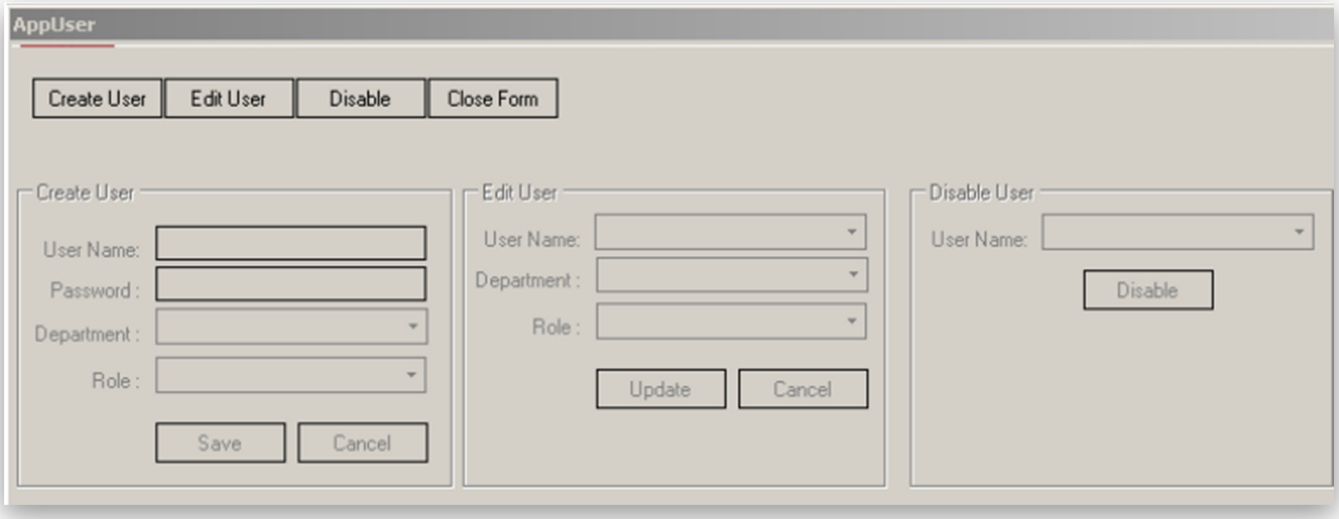
3. *Reconciliation Module*

Under the MIS Reporting module, Reconciliation module is helping banks to save their time and efforts to perform reconciliation using specific reconciliation reports.

Security Module

In Security module, system admin can create/edit and disable users as and when needed. Users can be created in following two predefined roles;

- Sys Admin
- Basic User



REQUIREMENTS FOR INFBOTS GATEWAY APPLICATION

Hardware Requirement

Component	Minimum Requirement
Computer and processor	A computer with an Intel Pentium-compatible CPU that is: • Core i7 6700 3.4GHZ **(6TH GENERATION)** Hyper-threading and dual-core processors are supported. Computers based on CPUs that are compatible with the AMD32 and AMD64 (x86-64) and Extended Memory 32/64-bit Technology processor architecture are considered x86/x64-based systems.
Memory	• 8 GB RAM or Higher
Hard disk	1 TB of available hard disk space for a complete installation including the operating system and all prerequisite software. The hard disk must be NTFS formatted.

Software Requirement

Microsoft Windows	Windows Server 2012, Windows Server 2008 R2 SP1, Windows 7 SP1, Windows 8.
Web Server (IIS)	Windows Internet Information Server(IIS) 7 or Higher
Microsoft .NET Framework	.NET Framework 4 or Higher
Database	SQL Server Standard Edition 2008 R2 or SQL Server Enterprise Edition 2008 R2
Adobe PDF Reader	Any Adobe PDF Reader

DELIVERABLES, PRICING & TERMS

Deliverables

Under the solution deployment agreement following are the deliverables against above mention scope of work;

- Software application installation package including installation & configuration manual
- Operating & Administration user manual
- At least one on-site technical resource deployment for minimum three working days at the time of software application Go-live.

Project Pricing

S#	Milestone	Price (PKR)
1	STP - RTGS Gateway Application -	2,000,000
2	Home Remittance - Annexure B to SWIFT Converter Utility	500,000
3	Operator Application & Integration with Core Banking App.	500,000
	TOTAL PROJECT COST & TIMEFRAME…….	3,000,000

Payment Schedule

S#	Milestone	Estimated Time	Due Amount (PKR)
	Agreement signing (Up-on Work Order)	-	1500,000
1	Phase – I Compliance of Home Remittance (March 31 2016)	2 – 3 Weeks	500,000
	Phase – II Deployment of Message Creation Application UAT Sign-off	6 – 8 Weeks	500,000
2	Phase – III Deployment of RCC Communication Channel in replacement of SBP Controller (Project Finalization) UAT Sing-off	4 - 6 Weeks	500,000

Proposal Signing Terms

1. Prices are net-off all taxes and valid till 31st of March, 2016.
2. JS Bank will provide dedicated business and technical resources to the project to facilitate our Consultants.
3. Invoices of charges are payable within 15 days of their being raised and will bear the date of their issuance.

WARRANTY & ONGOING SUPPORT

Warranty

- Warranty will be expired after one year from the date of project initiation.
- During warranty period, diagnosis and resolution of the problem/issues in the existing System be covered.
- Out of duly signed scope document, Requirement(s) raised by JS Bank will be cater through change management, which would be addressed through new proposal with pricing.

Ongoing Support

- Ongoing Service Contract for a minimum of 1 year.
- After warranty period expired, InfoBits will charge 20% of total project cost for Yearly Software Technical Support.
- Support contract will cover diagnosis and resolution of the problem/issues in the existing System
- Any mandatory change/modification in RTGS Gateway application directed by State Bank of Pakistan, will be covered under ongoing support contract
- A separate requisition would be submitted by JS Bank for new development and up gradation, which would be addressed through new proposal with pricing.
- INFOBITS will provide onsite, online, and telephonic support on a business day basis, Business day is defined as 9:00 AM through 7:00 PM Pakistan Time, Monday to Friday excluding holidays and weekends.

Out of Scope

Under the ongoing software technical support agreement, followings will remain out of scope;

- New Message Type introduced by either State Bank of Pakistan or JS Bank.
- New Input Interface on existing message types to manage internal controls.
- New management information report.